

MINGW64:/c/Users/Nitika/Nitika-GLA-LibrarySystem

```
$ git status
On branch main
nothing to commit, working tree clean
$ git log --oneline --all --decorate
8d45e10 (HEAD -> main) Resolve merge conflict in library_info.txt
6ab92f1 Main project information update
4c71e20 (library-update) Update library project information
3f82a16 (book-feature) Add Book Management Module
9d84b71 Add project objective and description
7a31c2d Initial project setup
$ git branch -a
  book-feature
  library-update
* main
$ tree
.
|-- backup
|-- config
|-- docs
|-- library_info.txt
`-- src
    |-- book.txt
$ cat library_info.txt
Name: Nitika Yadav
Roll Number: 2415001891
Subject: BCSE1522 - Application Build and Release Automation Lab
Project Name: Online Library Management System
Task 5 – Submission | Nitika Yadav | 2415001891
```

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```
$ git switch main
Already on 'main'
$ git merge book-feature
Updating 9d84b71..3f82a16
Fast-forward
 src/book.txt | 7 ++++++
 1 file changed, 7 insertions(+)
$ git switch -c library-update
Switched to a new branch 'library-update'
$ echo "Project Name: Library Management System - Library Update" > library_info.txt
$ git add library_info.txt
$ git commit -m "Update library project information"
[library-update 4c71e20] Update library project information
 1 file changed, 1 insertion(+), 6 deletions(-)
$ git switch main
Switched to branch 'main'
$ echo "Project Name: Online Library Management System - Main Update" > library_info.txt
$ git add library_info.txt
$ git commit -m "Update project information on main"
[main 6ab92f1] Update project information on main
 1 file changed, 1 insertion(+), 6 deletions(-)
$ git merge library-update
Auto-merging library_info.txt
CONFLICT (content): Merge conflict in library_info.txt
Automatic merge failed; fix conflicts and then commit the result.
$ git status
On branch main
You have unmerged paths.
  both modified:   library_info.txt
$ cat library_info.txt
<<<<<< HEAD
Project Name: Online Library Management System - Main Update
=====
Project Name: Library Management System - Library Update
Task 4 - Merge Conflict | Nitika Yadav | 2415001891
```

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```
$ git checkout -b book-feature
Switched to a new branch 'book-feature'
$ cat > src/book.txt
Book Management Module

Features:
1. Add and store book details
2. Search books by title or author
3. Update book availability status
4. Remove books from the library
$ git add src/book.txt
$ git commit -m "Add Book Management Module"
[book-feature 3f82a16] Add Book Management Module
 1 file changed, 7 insertions(+)
$ git branch
* book-feature
  main
$ git status
On branch book-feature
nothing to commit, working tree clean
$ git switch main
Switched to branch 'main'
$ git branch
  book-feature
* main
$ git branch -a
  book-feature

Task 3 – Feature Branch | Nitika Yadav | 2415001891
```

MINGW64:/c/Users/Nitika/Nitika-GLA-LibrarySystem

```
$ git init
Initialized empty Git repository in /c/Users/Nitika/Nitika-GLA-LibrarySystem/.git/
$ git config user.name "Nitika Yadav"
$ git config user.email "nitika.yadav_cs24@gla.ac.in"
$ git status
On branch main
No commits yet
Untracked files:
  backup/  config/  docs/  library_info.txt  src/
$ git add .
$ git status
Changes to be committed:
  new file:   library_info.txt
  new file:   src/
  new file:   docs/
  new file:   config/
  new file:   backup/
$ git commit -m "Initial project setup"
[main (root-commit) 7a31c2d] Initial project setup
 5 files changed, 5 insertions(+)
$ echo "Objective: Manage books, members and library records efficiently." >> library_info.txt
$ echo "Description: A simple Online Library Management System for GLA Digital Solutions." >> library_info.txt
$ git diff
+Objective: Manage books, members and library records efficiently.
+Description: A simple Online Library Management System for GLA Digital Solutions.
$ git add library_info.txt
$ git commit -m "Add project objective and description"
Task 2 – Git Repository | Nitika Yadav | 2415001891
```

MINGW64:/c/Users/Nitika

```
$ mkdir Nitika-GLA-LibrarySystem
$ cd Nitika-GLA-LibrarySystem
$ mkdir src docs config backup
$ echo "Name: Nitika Yadav" > library_info.txt
$ echo "Roll Number: 2415001891" >> library_info.txt
$ echo "Subject: BCSE1522 - Application Build and Release Automation Lab" >> library_info.txt
$ echo "Project Name: Online Library Management System" >> library_info.txt
$ echo "Semester: 5th Semester" >> library_info.txt
$ pwd
/c/Users/Nitika/Nitika-GLA-LibrarySystem
$ ls
backup  config  docs  library_info.txt  src
$ cat library_info.txt
Name: Nitika Yadav
Roll Number: 2415001891
Subject: BCSE1522 - Application Build and Release Automation Lab
Project Name: Online Library Management System
Semester: 5th Semester
$ tree
.
|-- backup
|-- config
|-- docs
|-- library_info.txt
`-- src
```

Task 1 – Project Workspace | Linux & Git Version Control